

Optimisation of a container [Cambridge insert]

One constraint, one variable, one derivative — and the cheapest tin in the shop.

LESSON 55–56

2 × 40 MIN

GEOMETRY 11, AMALIY MASALALAR

P1 9.3

LESSON CLOCK

15'

30'

25'

25'

10'

The four steps	15 min
The closed cylinder	30 min
Open containers	25 min
Cost, not area	25 min
Homework	10 min

LEARNING OBJECTIVES

- Form an expression for the quantity to be optimised.
- Use the constraint to reduce it to one variable.
- Differentiate, solve $f' = 0$ and verify the nature of the stationary point.
- Interpret the answer, including any practical restriction.

TEXTBOOK

National	pp. 291–300
Cambridge	Pure Mathematics 1, pp. 196–204
Workbook	Exercise 9C

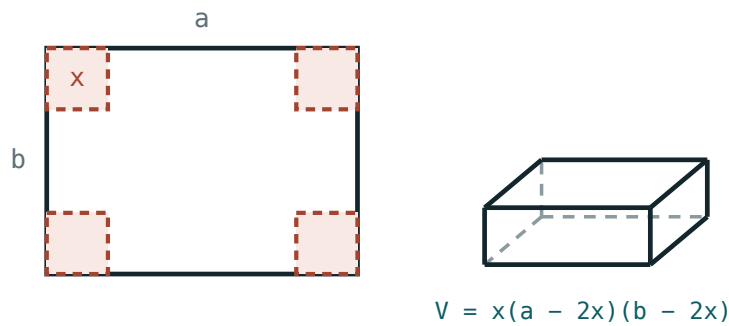
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Explanation

The four steps

EVERY OPTIMISATION QUESTION, IN FOUR STEPS

- Write the quantity to be optimised (the **objective**) in terms of the dimensions.
- Write the **constraint**, and use it to eliminate one variable.
- Differentiate, set $= 0$, solve.
- Verify it is a minimum (or maximum) and answer the question asked.



cut a square x from each corner

The classic box — one constraint, one variable, one derivative.

STEP 4 IS A REAL MARK

Finding r is not the answer if the question asked for the **area**, or for h , or for the ratio $\frac{h}{r}$. And a stationary point must be shown to be a minimum, by $f'' > 0$ or a sign change of f' .

The closed cylinder

The standard question: a closed cylindrical tin of fixed volume V — what shape uses least metal?

STEP	WORKING
objective	$S = 2\pi r^2 + 2\pi rh$
constraint	$V = \pi r^2 h \Rightarrow h = \frac{V}{\pi r^2}$
eliminate	$S = 2\pi r^2 + \frac{2V}{r}$
differentiate	$\frac{dS}{dr} = 4\pi r - \frac{2V}{r^2}$
solve	$4\pi r^3 = 2V \Rightarrow r = \sqrt[3]{\frac{V}{2\pi}}$
verify	$\frac{d^2S}{dr^2} = 4\pi + \frac{4V}{r^3} > 0 — a\ minimum$

THE ANSWER IN ONE SENTENCE

Substituting back gives $h = 2r$: **the least-metal closed tin is as tall as it is wide**. Its height equals its diameter, and it fits exactly inside a cube. The result does not depend on V at all.

Example. $V = 2000\text{ cm}^3$ gives $r = \sqrt[3]{\frac{2000}{2\pi}} \approx 6.83\text{ cm}$ and $h \approx 13.66\text{ cm}$, with $S \approx 879\text{ cm}^2$. A tin of the same volume with $r = 4$ needs $S \approx 1101\text{ cm}^2$ — a quarter more metal for the same contents.

Open containers

Remove a face and the objective changes, so the answer does too.

CONTAINER	OBJECTIVE	OPTIMAL SHAPE
closed cylinder	$2\pi r^2 + 2\pi rh$	$h = 2r$
open cylinder (no lid)	$\pi r^2 + 2\pi rh$	$h = r$
closed cuboid, square base	$2x^2 + 4xh$	$h = x$
open cuboid, square base	$x^2 + 4xh$	$h = \frac{x}{2}$

Every one is the same four steps. Note the pattern: removing the lid makes the optimal container **shallower**, because height is now relatively more expensive than base.

A CUBE IS OPTIMAL ONLY WHEN THE BOX IS CLOSED

The closed cuboid of least area for a given volume is a cube. The open one is **not** — it is half as tall as it is wide. Assuming a cube without checking is a common and costly shortcut.

Cost, not area

When materials differ in price, minimise the **cost**, not the area — the four steps are unchanged, but the objective carries the prices.

Example. An open tank with a square base of side x and height h holds 32 m^3 . The base costs 10 units per m^2 , the sides 5 :

$$C = 10x^2 + 5(4xh) = 10x^2 + 20xh, h = \frac{32}{x^2}$$

$$C = 10x^2 + \frac{640}{x}, \quad \frac{dC}{dx} = 20x - \frac{640}{x^2} = 0$$

So $x^3 = 32$, $x \approx 3.17$ m, $h \approx 3.17$ m — and here $h = x$, not $\frac{x}{2}$, because the base is twice as expensive and the design compensates by making it smaller.

CHECK THE FEASIBLE RANGE

A dimension must be positive, and often has an upper bound from the physical situation (a sheet of given size, a shelf of given height). If the stationary point lies outside that range, the optimum is at an **endpoint** — and the derivative never finds it. State the range before differentiating.

02 Worked examples

EXAMPLE 1 A closed cylindrical tin holds 2000 cm^3 . Find the radius and height that use least metal.

$$1 \quad S = 2\pi r^2 + \frac{4000}{r}$$

After eliminating h .

$$2 \quad \frac{dS}{dr} = 4\pi r - \frac{4000}{r^2} = 0$$

$$3 \quad r^3 = \frac{1000}{\pi} \Rightarrow r \approx 6.83$$

$$4 \quad h = \frac{2000}{\pi(46.6)} \approx 13.66 = 2r$$

$S'' > 0$, so a minimum.

ANSWER

$r \approx 6.83 \text{ cm}$, $h \approx 13.66 \text{ cm}$

EXAMPLE 2 An open cylindrical tank holds $54\pi \text{ cm}^3$. Find the radius that uses least metal.

$$(1) \quad S = \pi r^2 + 2\pi rh, \quad h = \frac{54}{r^2}$$

$$(2) \quad S = \pi r^2 + \frac{108\pi}{r}$$

$$(3) \quad \frac{dS}{dr} = 2\pi r - \frac{108\pi}{r^2} = 0 \Rightarrow r^3 = 54$$

$$(4) \quad r \approx 3.78, \quad h \approx 3.78 = r.$$

ANSWER

$$r = \sqrt[3]{54} \approx 3.78 \text{ cm, with } h = r$$

EXAMPLE 3 An open tank with a square base holds 32 m^3 . The base costs 10 and the sides 5 units per m^2 . Find the cheapest dimensions.

$$(1) \quad C = 10x^2 + 20xh, \quad h = \frac{32}{x^2}$$

$$(2) \quad C = 10x^2 + \frac{640}{x}$$

$$(3) \quad \frac{dC}{dx} = 20x - \frac{640}{x^2} = 0 \Rightarrow x^3 = 32$$

$$(4) \quad x \approx 3.17 \text{ m}, \quad h \approx 3.17 \text{ m}; \quad C \approx 302 \text{ units.}$$

ANSWER

$$x = h \approx 3.17 \text{ m}$$

EXAMPLE A rectangular sheet 40×40 cm has equal squares of side x cut from the corners; the flaps fold up.
 4 Find x for the greatest volume.

1 $V = x(40 - 2x)^2$, with $0 < x < 20$.

State the range.

2 $\frac{dV}{dx} = (40 - 2x)^2 - 4x(40 - 2x)$

3 $(40 - 2x)(40 - 6x) = 0 \Rightarrow x = 20$ (rejected) or $x = \frac{20}{3}$

4 $x \approx 6.67$ cm, $V \approx 4741$ cm³.

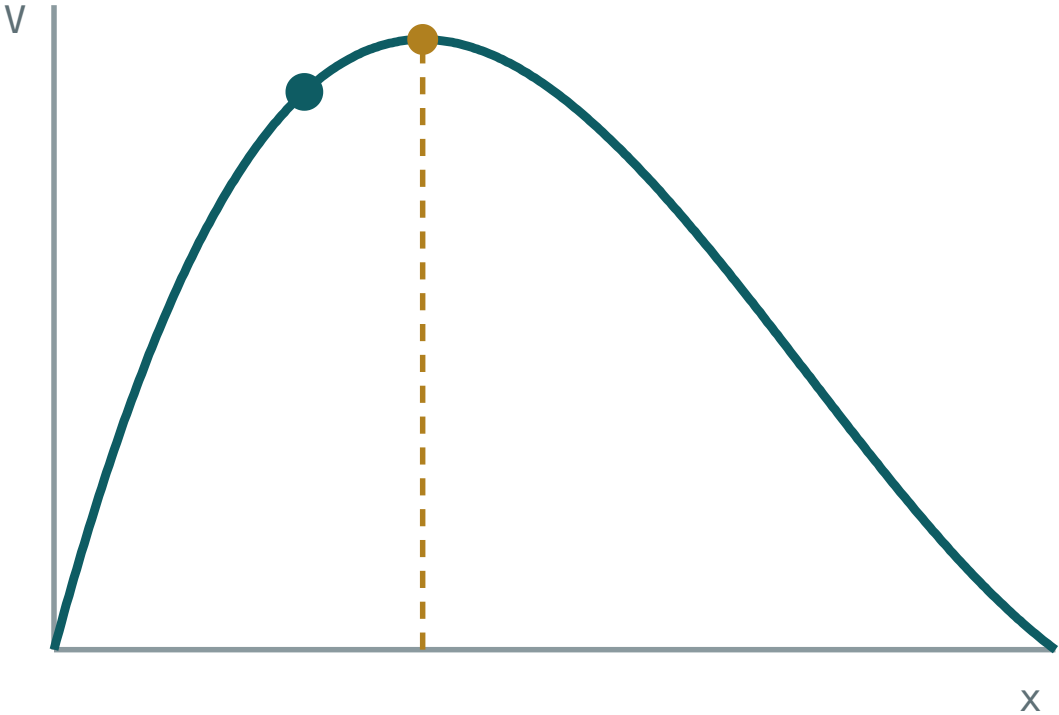
ANSWER

$$x = \frac{20}{3} \approx 6.67 \text{ cm}$$

03 Interactive model

Bring three tins of the same volume but different shapes; the class predicts which uses least metal, then measures.

The cheapest container



x 2	V(x) 384	greatest V 420.1	at x = 2.94
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Optimisation	Optimallashtirish	Оптимизация
Constraint	Cheklov	Ограничение
Objective function	Maqsad funksiyasi	Целевая функция
Stationary point	Statsionar nuqta	Стационарная точка
Minimum	Minimum	Минимум
Maximum	Maksimum	Максимум
Second derivative	Ikkinchi hosila	Вторая производная
Feasible range	Mumkin bo'lgan oraliq	Допустимый интервал
Cost function	Narx funksiyasi	Функция стоимости
Aspect ratio	Nisbat	Отношение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The constraint is used to:

check the answer

eliminate a variable

differentiate

nothing

A minimum is confirmed by:

$$f' = 0$$

$$f'' > 0$$

$$f'' < 0$$

$$f = 0$$

A closed cylinder of least area has:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 4r$$

An open cylinder of least area has:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 3r$$

A closed cuboid of least area is:

a cube

twice as tall as wide

half as tall as wide

any shape

An open cuboid of least area has:

$$h = x$$

$$h = \frac{x}{2}$$

$$h = 2x$$

$$h = 4x$$

If the stationary point is outside the feasible range:

there is no answer

the optimum is at an endpoint

differentiate again

ignore the range

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Objective for a closed cylinder

ANSWER $2\pi r^2 + 2\pi rh$

2. Objective for an open cylinder

ANSWER $\pi r^2 + 2\pi rh$

3. Constraint for a cylinder of volume V

ANSWER $h = \frac{V}{\pi r^2}$

4. Optimal shape of a closed cylinder

ANSWER $h = 2r$

5. Optimal shape of an open cylinder

ANSWER $h = r$

6. Optimal closed cuboid with square base

ANSWER $h = x$ — a cube

7. Test for a minimum

ANSWER $f'' > 0$

Medium · the standard question

7 PROBLEMS

1. Closed tin of 2000 cm^3 : the radius

ANSWER $\approx 6.83 \text{ cm}$

2. Same tin: the height

ANSWER $\approx 13.66 \text{ cm}$

3. Same tin: the least area

ANSWER $\approx 879 \text{ cm}^2$

4. Open tank of $54\pi \text{ cm}^3$: the radius

ANSWER $\sqrt[3]{54} \approx 3.78$

5. Open cuboid, square base, volume 108 : the base edge

ANSWER $x = 6$

6. Box from a 40×40 sheet: the cut x

ANSWER $\frac{20}{3} \approx 6.67$

7. Same box: the greatest volume

ANSWER $\approx 4741 \text{ cm}^3$

Hard · stretch and reason

7 PROBLEMS

1. Open tank, 32 m³, base 10 and sides 5 per m²: the base edge

ANSWER ≈ 3.17 m

2. Same: the least cost

ANSWER ≈ 302 units

3. A cylinder inscribed in a sphere of radius R : the greatest volume

ANSWER $\frac{4\pi R^3}{3\sqrt{3}}$

4. A cone of slant l : the greatest volume

ANSWER $h = \frac{l}{\sqrt{3}}$

5. A closed cylinder where the ends cost twice the side: the optimal $\frac{h}{r}$

ANSWER 4

6. A box from a 30 × 50 sheet: the cut x for greatest volume

ANSWER ≈ 6.07 cm

7. A cylinder of volume V inscribed in a cone of radius R and height H : the greatest V

ANSWER $\frac{4\pi R^2 H}{27}$

07 Homework

Homework — 5 tasks

Write the four steps as headings and fill them in; the structure carries the method marks.

1. A closed cylindrical tin holds 1000 cm³. Find the radius and height that use least metal, and verify the minimum.
2. An open cylindrical tank holds 16π m³. Find the radius that uses least material.
3. An open box with a square base holds 500 cm³. Find the dimensions of least surface area.

4. Squares of side x are cut from the corners of a 24×36 cm sheet and the flaps folded up. Find x for the greatest volume, stating the feasible range.
5. An open tank of 18 m^3 has a square base costing 8 and sides costing 4 units per m^2 . Find the cheapest dimensions.